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Development and application of a triplex-PCR assay for rapid detection of Methicillin resistant *Staphylococcus aureus* (MRSA) from pigs

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Running headline: Triplex-PCR for rapid detection of MRSA from pigs

Significance and Impact of the Study: The study reports the development and evaluation of a triplex-PCR assay for the rapid detection of methicillin-resistant *Staphylococcus aureus* (MRSA) from pigs. There is an urgent public health need for early and reliable detection of MRSA infection not only to reduce the cost of treatment but also to direct appropriate antibiotic therapy for its better management. This assay will be very useful for the rapid detection of MRSA from pigs.

Abstract

A triplex-PCR assay was developed and evaluated for rapid detection of methicillin-resistant *Staphylococcus aureus* (MRSA) recovered from various biological samples of pig. Three sets of primers were designed to target *mecA*, *16S* *rRNA* and *nuc* genes of MRSA. The specific amplification generated 3 bands on agarose gel, with sizes 280 bp for *mecA*, 654 bp for *16S rRNA* and 481 bp for *nuc*, respectively. A potential advantage of the PCR assay is its sensitivity with a detection limit of 10^2 CFU ml^{-1} of bacteria. Seventy nine MRSA isolates recovered from various samples of pigs were subjected to the amplification by the triplex- PCR assay and all the isolates yielded 3 bands corresponding to the three genes under this study. No false positive amplification was observed, indicating the high specificity of the developed triplex-PCR assay. This assay will be a useful and powerful method for differentiation of MRSA from methicillin-sensitive *S. aureus,* coagulase negative methicillin-resistant staphylococci and coagulase negative methicillin-sensitive staphylococci.

Keywords:

methicillin resistant *Staphylococcus aureus,* pig, sensitivity, specificity, triplex-PCR

Introduction

Staphylococcus aureus is a common bacterium found on the skin and nasal passages of healthy people. This bacterium is well known as colonizer of the human skin, and is a leading cause of diseases ranging from mild skin and soft tissue infections to life-threatening illnesses, such as deep post-surgical infections, septicaemia and toxic shock syndrome (Reddy *et al*. 2017). Methicillin-resistant *Staphylococcus aureus* (MRSA) is an *S. aureus* that became resistant to β-lactam antibiotics by acquiring *mecA* gene on its chromosomal DNA (Hiramatsu *et al*. 2013). The *mecA* gene is located on a mobile genetic element, designated as staphylococcal cassette chromosome *mec* (SCC *mec*), which contains the *mec* gene complex (the *mecA* gene and its regulators) and the *ccr* gene complex encoding site-specific recombinases responsible for the mobility of SCC*mec* (Lakhundi and Zhang 2018).

MRSA has become a pathogen of increasing importance in hospitals (healthcare-associated MRSA), the community (community-associated MRSA) and livestock operations (livestock-associated MRSA) (Graveland *et al*. 2011). To date, livestock-associated MRSA (LA-MRSA) has been found worldwide, particularly among people who are involved with livestock farming (Frana *et al*. 2013). These bacteria can be transmitted to humans in close contact with MRSA-colonized animals

(Smith and Pearson 2011). Livestock, especially pigs can serve as reservoirs for LA-MRSA (Sorensen *et al.* 2017). As pig can serve as reservoirs for LA-MRSA, timely detection of MRSA in pig population by a reliable diagnostic assay coupled with measures to prevent the spread of MRSA from pigs to human, especially those who are in close contact with pig can help in reducing MRSA burden in human.

Antibiotic resistance in microbes is one of the major global concerns in public health. MRSA represents an important group (resistant to several antibiotics) which is commonly referred to as "Super Bugs" (Xu *et al*. 2011). The prevention and control of multidrug-resistant MRSA spread remains a major challenge of infection-control practitioners worldwide (Khaled *et al*. 2019). With the emergence of glycopeptide-resistant *S. aureus,* control of MRSA is becoming even more essential (McGuinness *et al*. 2017). Recent studies have also reported the cost-effectiveness of strategies to prevent MRSA transmission and infection in intensive care unit (Gidengil *et al.* 2015). Current control strategies to limit the emergence and rapid spread of this organism rely on rapid and sensitive tests for detection of MRSA carriers.

Routine culture-based diagnostic procedure for potentially pathogenic *Staphylococcus* strains includes: enrichment in liquid media, isolation of colonies on selective agar plates followed by DNase or coagulase assays and confirmation by biochemical tests. However, the prolonged recovery time to identify microbes at the species level, labour intensiveness as well as material consuming, false negative results and insufficient sensitivity have raised concerns for these traditional methodologies (Franco-Duarte *et al*. 2019). During the past decades, a number of PCR and real-time quantitative PCR based assays have been employed for rapid detection of *Staphylococcus* (Ali *et al*. 2014; Velasco *et al*. 2014). However, limitations for simplex PCR (time required for detection of individual targeted gene of interest) and real-time PCR (requirement for trained personnel, operating space, expensive equipment and reagents) posed significant obstacles for their broad application. Considering the risk and hazard of MRSA strains, it is very much essential to develop a diagnostic assay which is capable of timely and sensitive detection of MRSA. The present study involves the development and evaluation of a triplex-PCR assay for rapid detection of MRSA recovered from various biological samples of pigs.

Results and discussion

The current report describes the development and evaluation of a triplex PCR assay for rapid detection of methicillin-resistant *S. aureus* (MRSA) obtained from various biological samples of pig. This developed assay will also help in differentiating MRSA from methicillin-sensitive *S. aureus,* coagulase negative methicillin-resistant staphylococci and coagulase negative methicillin-sensitive staphylococci. However, this PCR assay cannot detect the mecC that lead to methicillin resistance.

Multiplex PCR is an essential cost-saving technique for large scale scientific, clinical and commercial applications (Xu *et al*. 2012). Although simplex PCRs have been used for detection of MRSA from pigs (Pillai *et al*. 2012; Pournajaf *et al*. 2014) it cannot be used for simultaneous detection of multiple genes on interest.

Specificity of the primers in the present study was determined with different strains of Staphylococci, *Salmonella typhimurium* and *Escherichia coli.* PCR assay was carried out with a mixture of primers and DNA template prepared from respective strains. The results indicated that each primer pair was specific for the corresponding target genes (Fig. 1). The triplex -PCR assay was highly specific for the detection of target gene (*mecA*) of MRSA and no amplification products were obtained for strains of *Salmonella typhimurium* and *Escherichia coli* used in the study. The assay could also differentiate MRSA from other staphylococci (methicillin-sensitive *S. aureus,* coagulase negative methicillin-resistant staphylococci, coagulase negative methicillin-sensitive staphylococci). There was complete agreement between the results of simplex and triplex-PCRs for all the well characterized strains tested in the study indicating the high degree of specificity of the assay. Seventy nine MRSA isolates recovered from various samples of pigs were subjected to the amplification by the triplex-PCR assay and all the isolates yielded 3 bands corresponding to the three genes under this study. It was also observed that the triplex-PCR results were the same as that of monoplex PCR results. No false positive amplification was observed, indicating the high specificity of the developed triplex- PCR assay which further validates its use as an optimal tool for rapid detection of MRSA from pigs.

A potential advantage of the triplex-PCR developed in this study is that it has a high level of sensitivity. The experiments showed that a detection limit using the triplex-PCR was 10^2 CFU ml^{-1} for simultaneous detection of all the target genes of the organism. Although the infectious dose varies among pathogen types, it is generally believed that most bacterial pathogens are able to cause infection when more than 10^3 CFU ml^{-1} are ingested (US EPA 1992). Thus, the detection sensitivity of the triplex PCR assay described in this study is within the infectious dose of most pathogens. The triplex-PCR assay developed in this study was effective for the detection of the target genes of the organism. The results also demonstrated that amplification of target genes was efficient for producing 3 clear bands.

In conclusion, the developed triplex-PCR assay will be a useful tool for the rapid detection MRSA from pig.

Materials and methods

Bacterial strains and DNA preparations

Well-characterized MRSA strains (*mecA*, *nuc* and *16S rRNA* positive), *Salmonella typhimurium*, methicillin-sensitive *S. aureus,* coagulase negative methicillin-resistant staphylococci and coagulase negative methicillin-sensitive staphylococci maintained at the Animal Health Laboratory of National Research Centre on Pig, Indian Council of Agricultural Research, Rani, Guwahati, India and *Escherichia coli* ATCC25922 (*mecA*, *nuc* and *16S rRNA* negative) were used for the study. Four (4) strains from each class of Staphylococci were used for the optimization of the triplex PCR. These strains were subjected to evaluation and optimization of triplex-PCR assay. The optimized triplex-PCR assay was further performed on a total of 79 MRSA isolates obtained from different types of samples of pig (28 nasal, 14 skin and

37 pork samples). All isolates (79 nos) were confirmed both phenotypically and genotypically. Bacterial isolates were subcultured twice onto 5% sheep blood Columbia agar plates (Himedia, Mumbai, India) prior to DNA extraction. Genomic DNA was extracted by using DNeasy Blood and Tissue Kit (Qiagen, CA, USA).

Designing of oligonucleotide primers

All the oligonucleotide primers used in this study were synthesized by Imperial Life Science (P) Ltd. India. As *nuc* gene is the stable standard *Staphylococcus aureus* gene marker (Bogestam *et al.* 2018), it was incorporated in the PCR assay. The primers were designed using Primer 3 software. The sequences of these primers are shown in Table 1 along with their corresponding GenBank accession numbers and predicted product sizes. The primer pairs were designed to have a similar thermal melting temperature (Tm), to have minimal base pairing interactions with other primers in the reaction and to yield products that differed from each other by approximately 100bp, which could be resolved by agarose gel electrophoresis. The specificity of each of the 6 primers was initially confirmed using BLASTN and the GenBank sequence database.

Triplex-PCR and analysis of its products

The PCR reaction mixtures contained 6 primers at a concentration of 20 pmol each, 3µl of template, 10× PCR buffer, 2.0 mmol l^{-1} $MgCl_2$, 0.2mmol l^{-1} of each dNTPs and 1 U Taq DNA polymerase in a final volume of 25µl. Polymerase chain reaction assays were done using a GeneAmp PCR system 9700 thermal cycler (Applied Biosystems, USA) with the following amplification conditions: an initial denaturation at 94ºC for 1 min, followed by 30 cycles of denaturation for 30 s at 94ºC, annealing at 55ºC for 30 s, extension for 30 s at 72ºC with the final extension for 7 min at 72ºC. Amplified PCR products were analyzed by gel electrophoresis in 2% agarose containing ethidium bromide (0.5 µg ml^{-1}). The products were visualized with UV illumination and imaged with gel documentation system (Alpha Infotech Corporation, Multi Image System, San Leandro, CA, USA).

In parallel, all the samples were also tested using standard conventional microbiology procedures in order to not only confirm the method of m-PCR system but also evaluate its performance.

Sensitivity and specificity assays

The sensitivity of the triplex -PCR was tested using serial dilution of bacterial culture prepared from overnight cultures. The organism was decimally diluted (10^9, 10^8, 10^7, 10^6, 10^5, 10^4, 10^3, 10^2, 10,<10 CFU ml^{-1}) in sterile saline. DNA was extracted from 1 mL of each dilution and subsequently tested using the triplex-PCR method. Each sampling experiment consisted of triplicate tests with two replicates (each experiment with two replicates at different time intervals).

The triplex-PCR specificity was determined by examining the ability of the test to detect and distinguish used bacterial strains in the study.

Application of the triplex-PCR assay on clinical isolates

A total of 79 MRSA isolates obtained from different types of samples of pig (28 nasal, 14 skin and 37 pork samples) during the period of 2014-2019 were used. DNA from these isolates was extracted as mentioned above and was stored at -80°C until use. The DNA samples from these isolates were tested by the triplex-PCR assay. These experiments were replicated to ensure reproducibility.

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Conflict of Interest

There is no conflict of interest for this manuscript.

References

Ali, R. AI-Achkar, K., AI-Mariri, A. and Safi, M. (2014) Role of Polymerase Chain Reaction (PCR) in the detection of antibiotic-resistant *Staphylococcus aureus. Egypt* J Med Hum *Genet* **15**, 293-298.

Bogestam, K., Vondracek, M., Karlsson, M., Fang, H. and Giske, C.G. (2018)

Introduction of a hydrolysis probe PCR assay for high-throughput screening of methicillin-resistant *Staphylococcus aureus* with the ability to include or exclude detection of *Staphylococcus argenteus*. PLoS One **13**, e0192782.

Frana, T.S., Beahm, A.R., Hanson, B.M., Kinyon, J.M., Layman, L.L., Karriker, L.A., Ramirez, A. and Smith, T.C. (2013) Isolation and characterization of methicillin-resistant *Staphylococcus aureus* from pork farms and visiting veterinary students. *PLoS One* **8**, e53738.

Franco-Duarte, R., Černáková, L., Kadam, S., Kaushik, K. S., Salehi, B., Bevilacqua, A., Corbo, M. R., Antolak, H., Dybka-Stępień, K., Leszczewicz, M., Tintino, S. R., de Souza, V. C. A., Sharifi-Rad, J., Coutinho, H. D. M., Martins, N. and Rodrigues, C. F. (2019) Advances in Chemical and Biological Methods to Identify Microorganisms—From Past to Present. *Microorganisms* **7**, 130.

Gidengil, C., Gay, C., Huang, S. S. Platt, R. Yokoe, D. and Lee, G. M. (2015) Cost-effectiveness of strategies to prevent MRSA transmission and infection in an intensive care unit. *Infect Control Hosp Epidemiol* **36**, 17–27.

Graveland, H., Wagenaar, J.A., Bergs, K., Heesterbeek, H. and Heederik, D. (2011) Persistence of livestock associated MRSA CC398 in humans is dependent on intensity of animal contact. *PLoS One* **6**, e16830.

Hiramatsu, K., Ito, T., Tsubakishita, S., Sasaki, T., Takeuchi, F., Morimoto, Y., Katayama, Y., Matsuo, M., Kuwahara-Arai, K., Hishinuma, T. and Baba, T. (2013) Genomic Basis for Methicillin Resistance in *Staphylococcus aureus*. *Infect Chemother* **45**, 117–136.

Khaled, A., Moein, A. and Zaher, H. M. (2019) Occurrence of multidrug-resistant methicillin-resistant *Staphylococcus aureus* among healthy farm animals: a public health concern. *Int J Vet Sci Med* **7**, 55–60.

Lakhundi, S. and Zhang, K. (2018) Methicillin-Resistant *Staphylococcus aureus*: Molecular Characterization, Evolution and Epidemiology. *Clin Microbiol Rev* **31**, e00020-18.

McGuinness, W.A., Malachowa, N. and DeLeo, F. R. (2017) Vancomycin Resistance in *Staphylococcus aureus*. *Yale J Biol Med* **90**, 269–281.

Pillai, M. M., Latha, R. and Sarkar, G. (2012). Detection of Methicillin

Resistance in *Staphylococcus aureus* by Polymerase Chain Reaction and Conventional Methods: A Comparative Study. *J Lab Physicians* **4**, 83–88.

Pournajaf, A., Ardebili, A., Goudarzi, L., Khodabandeh, M., Narimani, T. and Abbaszadeh, H. (2014) PCR-based identification of methicillin-resistant *Staphylococcus aureus* strains and their antibiotic resistance profiles. *Asian Pac J Trop Biomed* **4**, S293–S297.

Reddy, P. N., Srirama, K. and Dirisala, V. R. (2017) An Update on Clinical Burden, Diagnostic Tools, and Therapeutic Options of *Staphylococcus aureus. Infect Dis (Auckl)*. **10**, 1179916117703999.

Smith, T.C. and Pearson, N. (2011) The emergence of *Staphylococcus aureus* ST398. *Vector Borne Zoon Dis* **11**, 327-339.

Sorensen, A.I.V., Toft, N., Boklund, A., Espinosa-Gongora, C., Græsboll, K., Larsen, J. and Halasa T. (2017) A mechanistic model for spread of livestock-associated methicillin-resistant *Staphylococcus aureus* (LA-MRSA) within a pig herd. *PLoS One* **12**, e0188429.

US EPA. (1992) Guidelines for water reuse. United States Environmental Protection Agency, Washington.

Velasco, V., Sherwood, J.S., Rojas-García, P. P. and Logue, C. M. (2014) Multiplex Real-Time PCR for Detection of *Staphylococcus aureus*, *mecA* and Panton-Valentine Leukocidin (PVL) Genes from Selective Enrichments from Animals and Retail Meat. *PLoS One* **9**, e97617.

Xu, W., Zhai, Z., Huang, K., Zhang, N., Yuan, Y., Shang, Y. and Luo, Y. (2012) A Novel Universal Primer-Multiplex-PCR Method with Sequencing Gel Electrophoresis Analysis. *PLoS One* **7**, e22900.

Xu, Z., Li, L., Shirtliff, M.E., Peters, B.M., Li, B., Peng, Y., Alam, M.J., Yamasaki, S. and Shi, L. (2011) Resistance class 1 integron in clinical methicillin-resistant *Staphylococcus aureus* strains in Southern China, 2001-2006. *Clin Microbiol Infect* **17**, 714-717.

Table 1 Oligonucleotide sequences used for triplex-PCR

Target genes	Oligonucleotide sequences	Amplicon size (bp)	GenBank Accession number
16S rRNA	F: TACATGCAAGTCGAGCGAAC R: CATTTCACCGCTACACATGG	654	DQ630753.1
nuc	F: GAAAGGGCAATACGCAAAGA R: AGCCAAGCCTTGACGAACTA	481	CP030661.1
mecA	F: AACGTTGTAACCACCCCAAG R: CCACCCTCAAACAGGTGAAT	280	CP030595.1

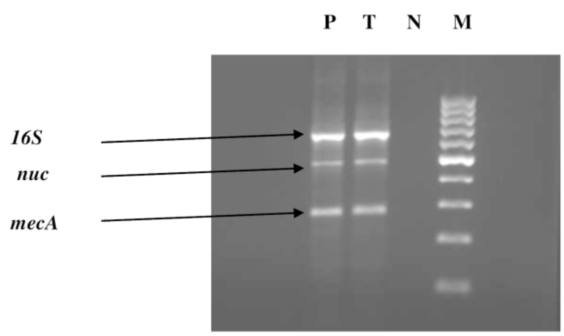

Figure 1 Triplex-PCR for simultaneous detection of *16S rRNA* (654bp), *mecA* (280bp) and *nuc* (481bp) genes of MRSA from pigs. Lane M: Molecular marker (100bp DNA ladder), Lane N: Negative control, Lane P: Positive control and Lane T: Test organism

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